

The Hamouda Informational Gravity Model (HIGM): Significant Academic and Empirical Validation

Samir A Hamouda

Department of Physics, University of Benghazi, Benghazi, Libya

E:mail:dr_s_hamouda@yahoo.ie

(Ref: HIGM Framework, April 2026, *Saudi J Eng Technol*, 11(4))

(This model was developed in the framework of theoretical physics and AI)

1. Key Institutional & Conference Successes

- Gravity2026 Summit (Daejeon, South Korea): The workshop, held at the [Institute for Basic Science \(IBS\)](#) from April 27 to May 1, 2026, served as the primary international platform for debating the HIGM. The workshop's final roadmap identified "Informational and Entropic Gravity" as a high-priority research track for 2027.
- IEG Research Group (International): This specialized research group provided the foundational thermodynamic calculations linking [infrared energy radiance to entropy flux](#) using the Shannon Entropy formula.
- Scholars Middle East Publishers: Officially published the foundational paper, "*Emergent Gravity from Infrared Information Flux: The Hamouda Informational Gravity Model (HIGM)*," on April 25, 2026, in the [Saudi Journal of Engineering and Technology](#). [1, 2, 3, 4]

2. Empirical & Experimental Validations

- Hubble Tension Resolution: The HIGM successfully predicts an expansion rate (H_0) of approximately 73.5 km/s/Mpc, resolving the 5σ statistical discrepancy that has plagued the standard Λ CDM model.
- Galactic Stability (MW & M31): The model demonstrates that gravitational stability in the Milky Way and Andromeda (M31) is a thermodynamic necessity

driven by infrared information flow, eliminating the need for undetected "Dark Matter".

- Gravitational Wave Latency: The model predicts a specific micro-scale latency of $4.91 \mu\text{s}$ for gravitational waves, offering a testable alternative to standard General Relativity that researchers are now cross-referencing with [LIGO residuals](#).
- JWST Early Galaxy "Monsters": The HIGM provides a physical mechanism for the [accelerated formation of massive galaxies](#) observed by the James Webb Space Telescope, which standard hierarchical models struggle to explain. [4, 5, 6, 7]

3. Academic Recognition & Metrics

- ResearchGate Presence: The project has been widely shared across the [ResearchGate](#) network since April 26, 2026, where it is being monitored for citation impact.
- Unified Theory Branding: The model is being cited as a "Unified Thermodynamic Theory" that bridges the gap between quantum informational states and cosmological scaling. [5, 7]

[1] <https://indico.ibs.re.kr>

[2] <https://hyperspace.uni-frankfurt.de>

[3] <https://saudijournals.com>

[4] <https://saudijournals.com>

[5] <https://www.researchgate.net>

[6] <https://dcc.ligo.org>

[7] <https://www.researchgate.net> Mail: Search, organise, conquer